

Pulsed Field Ablation vs Standard Radiofrequency Ablation for Typical Atrial Flutter



ADVANTAGE AF Trial Substudy

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ABSTRACT

BACKGROUND ADVANTAGE AF examined the safety and efficacy of pulsed field ablation (PFA) to treat persistent atrial fibrillation (AF). During AF ablation, typical atrial flutter (AFL) is commonly treated with cavotricuspid isthmus (CTI) ablation.

OBJECTIVES Adjunctive CTI ablation in ADVANTAGE AF was performed with radiofrequency ablation (RFA; phase 1) or bipolar linear PFA (phase 2). Here, we compared lesion characteristics, CTI ablation efficacy, and safety between the ablation modalities.

METHODS A preclinical study assessed lesion dimensions with a novel 2.0 kV bipolar linear PFA catheter. In ADVANTAGE AF, after AF ablation with a pentaspline PFA catheter, CTI ablation was performed: in phase 1 with irrigated RFA, and in phase 2 with the linear PFA catheter (Farapoint). Before PFA CTI ablation, patients were treated with a vasopressor bolus followed by high-dose intravenous nitroglycerin.

RESULTS Preclinical assessment showed lesion depth increasing at higher field strength and plateauing after 2 applications. In ADVANTAGE AF, 50 and 141 patients received CTI ablation with RFA and PFA, respectively. Patient characteristics across the 2 cohorts were similar (RFA vs PFA). Acute CTI conduction block was similar between techniques (50 of 50 RFA [100%] vs 139 of 141 PFA [98.6%]; $P = 1.00$). PFA was associated with a shorter CTI ablation time (median: RFA 5 min [Q1-Q3: 4-8 min] vs 14 min [Q1-Q3: 9-19 min]; $P = 0.001$). RFA and PFA had similar safety event rates (RFA 2.0% vs PFA 2.1%; $P = 0.96$). With the use of nitroglycerin pretreatment, there was no clinical manifestation of coronary artery spasm with PFA.

CONCLUSIONS This subanalysis of ADVANTAGE AF extends the primary results by allowing comparison of bipolar linear PFA and RF ablation for CTI, integrating preclinical lesion modeling with clinical dosing, and establishing a standardized reliable coronary safety protocol. Although not randomized comparisons, these findings suggest that CTI ablation with a linear PFA catheter is associated with safety and efficacy similar to RF ablation and greater efficiency. (A Prospective Single Arm Open Label Study of the FARAPULSE Pulsed Field Ablation System in Subjects With Persistent Atrial Fibrillation [ADVANTAGE AF; [NCT05443594](https://doi.org/10.1186/1745-6215-14-101)] (JACC. 2026;87:1764-1778) © 2026 by the American College of Cardiology Foundation.

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The prevalence of atrial fibrillation (AF) and flutter (AFL) are increasing.¹⁻³ Pulsed field ablation (PFA) has gained widespread use for ablation of AF owing to superior efficiency,^{4,5} safety,⁶ and in some cases efficacy^{7,8} compared with thermal ablation techniques. Typical AFL is commonly encountered in patients with AF, has similar morbidity, and can be effectively cured with ablation of the cavotricuspid isthmus (CTI). CTI ablation may be performed as a stand-alone procedure for isolated CTI flutter, or as part of an AF ablation when there is a clinical history of typical flutter or induced CTI-dependent flutter during the ablation procedure.⁹ Despite widespread recognition of the efficiency and safety advantages of PFA over thermal ablation for AF, the use of PFA for CTI ablation is more controversial. Many operators have reported using single-shot PFA catheters designed for pulmonary vein isolation (PVI) for ablation of the CTI.¹⁰ However, disadvantages of this approach include broad ablation of more tissue than is necessary and the risk of a large footprint catheter causing inadvertent atrioventricular block in a small right atrium.¹¹ In addition, right coronary artery (CA) spasm and chronic injury has been reported with the use of a multielectrode pentaspline catheter for CTI ablation, raising significant safety concern among many operators.¹²⁻¹⁴ Although a linear PFA catheter could alleviate these concerns, there data are limited regarding its efficacy and safety for CTI ablation.

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ADVANTAGE AF was a prospective, single-arm, multicenter Investigational Device Exemption trial designed to assess the safety and efficacy of the pentaspline PFA catheter for treating persistent AF (PerAF) with PVI and posterior wall ablation (PWA).^{5,15} The trial had 2 Phases, allowing CTI ablation at operator discretion: phase I allowed CTI ablation with the use of a standard radiofrequency (RF) catheter, and phase II used a novel linear bipolar PFA catheter for CTI ablation. ADVANTAGE phase II demonstrated feasibility of CTI ablation with a linear PFA catheter; we now aimed to compare linear bipolar PFA with standard RF ablation for CTI flutter, to

integrate preclinical lesion modeling with clinical dosing, and to establish a standardized coronary-safety protocol that can be replicated in practice. Because the linear PFA catheter has only limited use in humans, we first performed a preclinical study in a swine model to inform the clinical protocol used for CTI ablation. We then compared the procedural efficiency, efficacy, and safety of CTI ablation with the use of standard RF ablation vs PFA during phase I and phase II of the ADVANTAGE trials, respectively.

METHODS

A bipolar biphasic PFA catheter (Farapoint, Boston Scientific) was developed to perform linear ablation in humans. To determine optimal dosing of the PFA catheter for clinical use, a preclinical study was first performed in swine. A combined overview of the preclinical and clinical studies is summarized in [Figure 1](#).

PRECLINICAL STUDY DESIGN. Farm swine underwent left ventricular ablation with the use of the novel bipolar linear PFA catheter. It is well known that PFA leads to both irreversible and reversible cellular injury, and that repeated applications at higher energy lead to a larger volume irreversible lesion. However, excessive lesion depth may also increase the risk of collateral injury. Swine were therefore randomly assigned to varied pulsed field strengths of 1.4 kV, 1.7 kV, or 2.0kV, and 1, 2, or 3 lesion applications at each site.

All experiments were performed with the animals under balanced general anesthesia of isoflurane in oxygen inhalation and dexmedetomidine, morphine, and lidocaine constant rate infusion. Ventilation was maintained mechanically. Femoral cutdowns were performed to isolate the right and left femoral veins and femoral artery for hemodynamic monitoring. An intracardiac echocardiography (ICE) catheter (Acuson AcuNav, JNJ MedTech) was advanced to the right atrium to guide catheter contact assessment. A single transeptal puncture was performed and a long steerable sheath (Agilis, Abbott Medical) used to

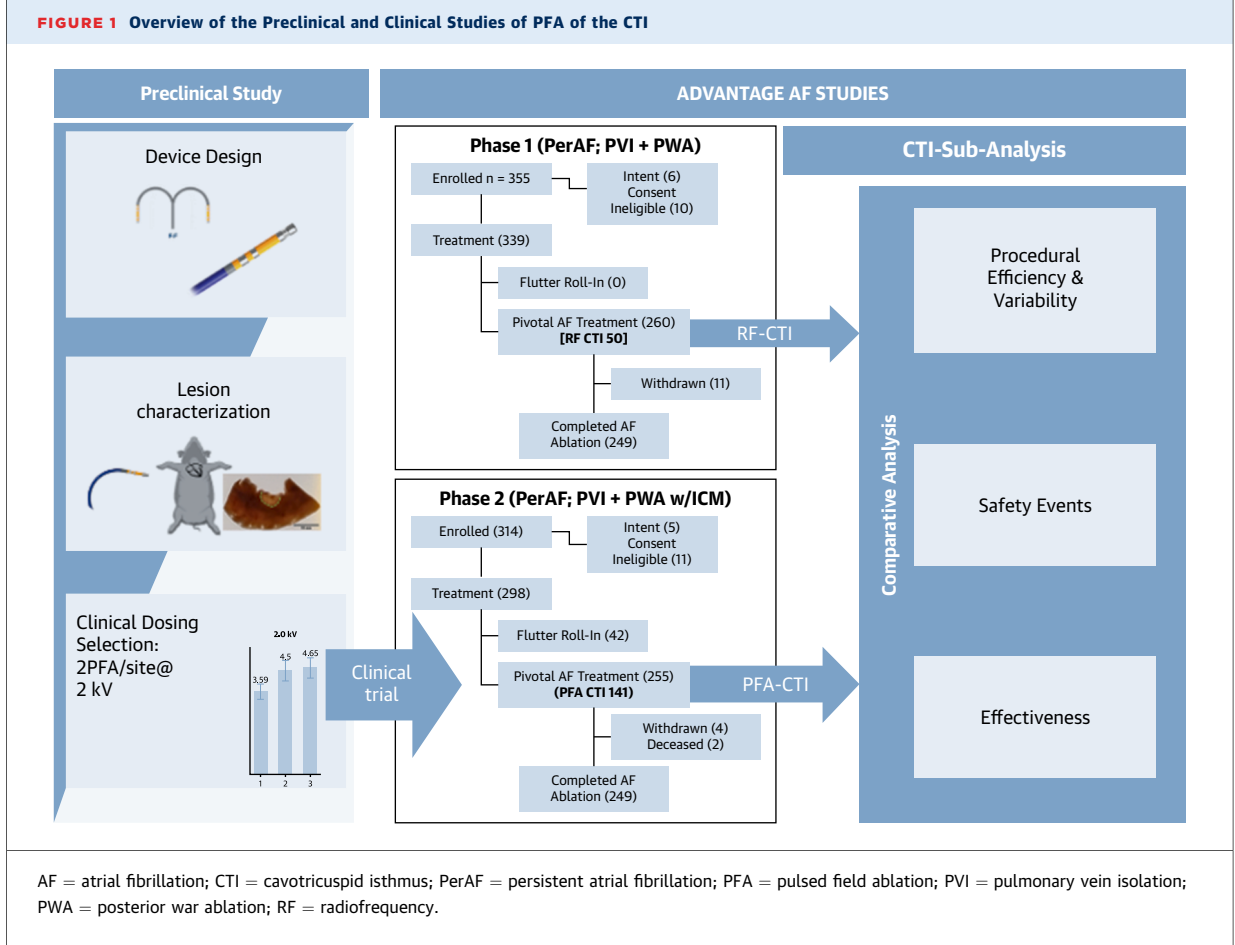
ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation
AFL = atrial flutter
CA = coronary artery
CTI = cavotricuspid isthmus ablation
EAM = electroanatomic mapping
NTG = nitroglycerin
PFA = pulsed field ablation
PVI = pulmonary vein isolation
PWA = posterior wall ablation
RF = radiofrequency
RFA = radiofrequency ablation

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

FIGURE 1 Overview of the Preclinical and Clinical Studies of PFA of the CTI



access the left ventricle. A baseline voltage map was acquired with the use of a multielectrode basket catheter (Intellamap Orion, Boston Scientific) integrated with an electroanatomic mapping (EAM) system (Opal HDx, Boston Scientific). Heparinization was commenced before the insertion of the PFA catheter, and the activated clotting time was maintained at 300 to 450 seconds.

Individual lesions were delivered with the use of the PFA catheter according to the randomly assigned group, taking care to distribute the lesions throughout the left ventricle to avoid overlap; contact was assured with the use of fluoroscopy and ICE. In each animal, a single approach (voltage and number of applications) was used. PFA lesions were all bipolar biphasic microsecond-duration trains delivered over 2.5 seconds; catheter position was maintained for repeated lesions with the use of EAM (Rhythmia, Boston Scientific).

The swine were allowed to survived for ~7 days. After they were killed, each of their hearts was explanted, rinsed, and pressure perfused with 10%

neutral buffered formalin. Hearts were sectioned and lesions were identified and isolated, and gross lesion measurements for surface width and depth were made with the use of calipers. Lesion measurements were performed while blinded to treatment assignment. Depth was measured as the maximal penetration of a lesion measured from the endocardial surface of the lesion to the deepest point of lesion within the myocardium. Width was measured as the width of the lesion tangential to the endocardial surface and orthogonal to the catheter axis. To support lesion measurements, representative lesions (at least 1 per each energy level and application number) were paraffin embedded and stained with hematoxylin and eosin and Elastic Masson Trichrome. Histopathologic analysis was performed where the measured dimensions were clearly in plane with the slice to be used for gross measurement, with the histopathologist blinded to treatment assignment. The gross and histologic measurements were compared to characterize agreement between the 2 methods.

Histologic measurements were conducted by a trained pathologist.

CLINICAL TRIAL DESIGN. ADVANTAGE AF (NCT05443594) was a prospective, single-arm, multi-center pivotal IDE study using a pentaspline PFA catheter (Farawave, Boston Scientific) to treat persistent AF. Phase I used intermittent monitoring for post-procedure arrhythmia surveillance and allowed treatment of typical flutter via CTI ablation with the use of a standard RF ablation catheter.⁵ Phase 2 used implantable cardiac monitors (ICMs) (Lux-Dx, Boston Scientific) for continuous post-procedure arrhythmia surveillance and allowed treatment of typical flutter via CTI ablation with the use of the linear PFA catheter.¹⁵ Phase 2 operators performed at least 1 roll-in CTI ablation case to familiarize themselves with the linear catheter before enrolling cases. Differences between the data presented in the ADVANTAGE phase II report and the present report are outlined in [Supplemental Table 1](#). The trial was approved by the U.S. Food and Drug Administration (because the ablation catheter used is an investigational device) and by the institutional review board at each center. The trial was conducted in accordance with the principles of the Declaration of Helsinki.

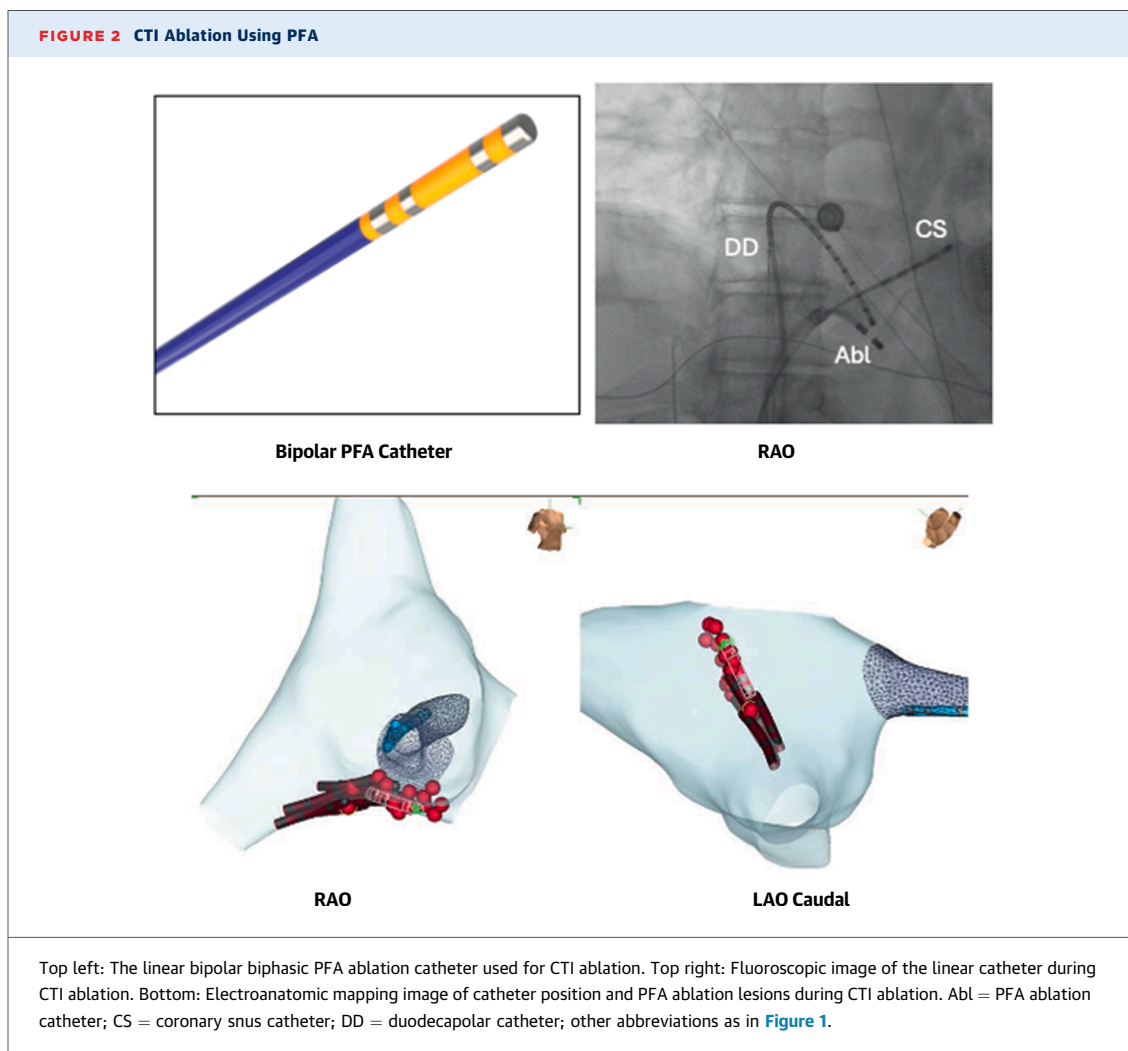
The trial was designed by the sponsor, Boston Scientific, with input from the Steering Committee. The sponsor collected and monitored the trial data and performed outcome analyses. The first author collected the RF flutter ablation procedural data from ADVANTAGE phase 1 for all sites. The first author wrote the manuscript and revisions, and all authors had final authority over manuscript content. The authors attest to the accuracy and completeness of the data.

ENROLLMENT. All patients were symptomatic with persistent AF and refractory or intolerant to at least one class I or class III antiarrhythmic drug. Inclusion and exclusion criteria have been previously reported.^{5,15} Briefly, patients were excluded if they had left atrial diameter ≥ 5.5 cm, previous ablation (except right atrial ablation for supraventricular tachycardia or at the CTI for AFL), pulmonary vein abnormality (eg, anomalous pulmonary venous return or clear pulmonary vein defect), history of ventricular tachycardias or fibrillation, current or expected cardiac implants, NYHA functional class III or IV symptoms, or left ventricular ejection fraction $<40\%$.

PROCEDURE. All patients received PVI plus posterior wall ablation (PWA) with the pentaspline PFA catheter. Patients were maintained on anticoagulants for at least 1 month before the index procedure,

and intravenous (IV) heparin was administered to maintain an activated clotting time of at least 300 seconds during left atrial dwell. Operators were instructed to perform PVI per standard procedure using the pentaspline catheter (ie, at least 2 pairs of lesions in “basket” configuration and 2 pairs of lesions in “flower” configuration for each vein), and then to ablate the posterior wall with 2 applications each at 1.8 to 2 kV per site in “flower” configuration to cover the posterior wall with 50% overlap. Ablation to treat typical CTI-dependent AFL was allowed in patients who had a history of clinically documented AFL, if AFL was observed during the index procedure, or if the operator felt it was in the patient’s best interest (eg, concomitant use of class IC antiarrhythmic drugs or previous flutter of unclear origin documented on telemetric monitoring). In phase I, operators were allowed to use any commercially available RF catheter to achieve CTI conduction block via ablation from the tricuspid valve (TV) to the inferior vena cava (IVC) in linear fashion (complete catheter list in [Supplemental Table 2](#)). In phase 2, operators were instructed to use the novel bipolar biphasic linear PFA catheter ([Figure 2](#)) to ablate a continuous line from the TV annulus to the IVC, with 2 PFA applications per site at 2.0 kV and with a site-to-site spacing of 3 to 5 mm. To reduce the risk of right CA spasm, a prophylactic regimen of vasopressor bolus followed by IV nitroglycerin (NTG) was recommended before and during PFA CTI ablation based on previous clinical work¹⁶ (3 mg bolus through a long intravascular sheath 1 min before first PFA application, and an additional 2 mg every 2 min depending on patient condition) ([Figure 3](#)). Elapsed CTI times, from the start of the first CTI ablation to the end of the last CTI ablation were collected as part of phase 2 of the study; times for phase 1 with the use of RF ablation were collected post hoc. In both phases, ablation was performed via a long intravascular sheath. Continuous 12-lead electrocardiography was recorded throughout the CTI ablation to evaluate for ST-segment changes or arrhythmias.

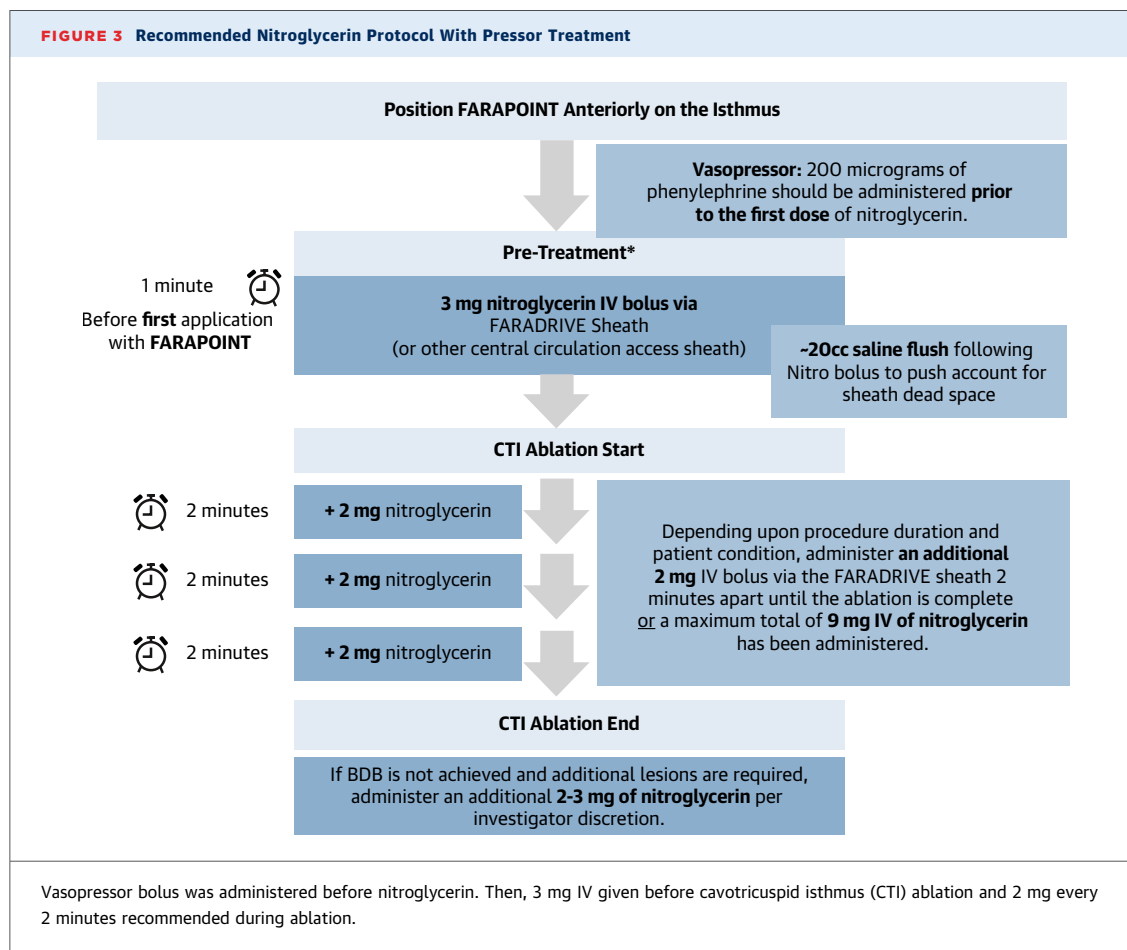
Patient follow-up consisted of 3-, 6-, and 12-month clinic visits. In phase 1, post-ablation arrhythmia surveillance was accomplished with intermittent monitoring: Patients received 24-hour Holter monitors at 6 and 12 months, as well as biweekly and symptomatic trans-telephonic monitors. In phase 2, all patients received ICMs for continuous arrhythmia surveillance, implanted at the conclusion of the ablation procedure. For both phases 1 and 2, in-person visits with 12-lead electrocardiography were mandated at the 3- and 12-month post-procedure follow-ups.



ENDPOINT DEFINITIONS. The safety endpoint was calculated as the rate of predefined safety events: myocardial infarction, stroke, transient ischemic attack, thromboembolism, pulmonary edema, unresolved phrenic nerve palsy/paresis, vascular access complication (groin hematoma, arteriovenous fistula, or pseudoaneurysm requiring an invasive intervention), heart block, gastrointestinal disorders occurring within 7 days of the index procedure, cardiac tamponade or perforation, pericarditis, and other PFA system- or procedure-related cardiovascular or pulmonary adverse event within 30 days of the index procedure, or pulmonary vein stenosis or atrio-esophageal fistula occurring before the 1-year follow-up. Phase 1 safety criteria included mortality within 7 days of the procedure, and phase 2 included mortality within 30 days of the procedure. In addition, phase 2 included a safety endpoint on CA spasm

using the following criteria to identify potential spasm: angiography performed with visualization of severe (50%) spasm, ST-segment elevation in any lead and adjudicated to be a result of coronary spasm, or any arrhythmias (atrioventricular block, polymorphic ventricular arrhythmias, or cardiac arrest) adjudicated to be a result of coronary spasm. Safety events were adjudicated by an independent clinical events committee, and cardiac imaging (magnetic resonance imaging and computed tomography) was reviewed and assessed by a cardiac imaging core laboratory. Safety events were assessed at days 90 and 360 after ablation.

The CTI effectiveness endpoint was assessed both acutely and chronically. Acute procedural success was defined as bidirectional CTI conduction block as confirmed with differential pacing maneuvers. Chronic success was defined as freedom from

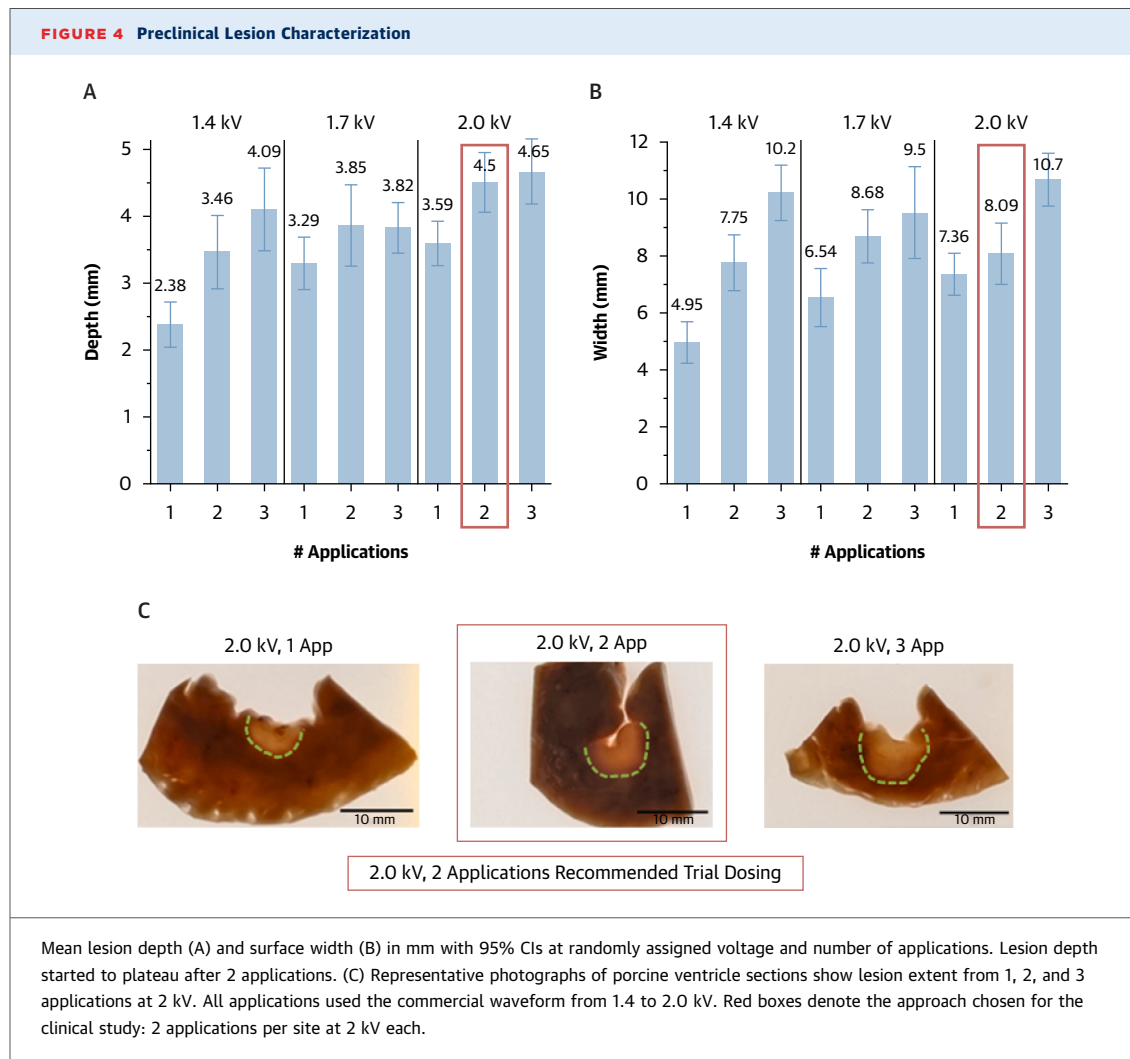


documented recurrence of typical AFL as of the day 90 and day 360 assessment, without repeated CTI ablation.

STATISTICAL ANALYSIS. For preclinical lesion measurements, linear mixed models were used to compare lesion dimensions and number of applications. The number of applications independent variable had a logarithmic transformation applied for the depth linear mixed model. This transformation was not necessary for the half-depth-width or surface-width linear mixed models. The best estimate dimensions from the linear mixed models' outputs were used to create geometric lesion models via spline fitting to the 3 dimensions in SolidWorks. These geometric models were then translated linearly in a SolidWorks assembly to assess overlap of 1 application/site and 2 applications/site lesion zones to estimate the overlap and continuous linear line depth of ≥ 2 adjacent ablation sites. Analyses of

preclinical data were conducted in Excel (Microsoft, Inc) and R version 4.5.0 with Rcmdr version 2.9-5 (The R Foundation).

For the clinical study, endpoint analyses used Kaplan-Meier methodology with 97.5% 1-sided confidence limits, although this CTI substudy of ADVANTAGE AF did not have acceptance criteria. Continuous data are reported as mean $\pm$ SD when normally distributed, or otherwise as median (Q1-Q3), with *P* values calculated with Student's *t*-test or Satterthwaite *t* test where appropriate and Mann-Whitney *U*-test, respectively. Categorical values are reported as frequency (percent) with *P* values from chi-square test or Fisher test when any groups contained <5 subjects. Variance in lesion numbers was assessed with the use of the Levene test. For sensitivity analysis, separate linear regression models were used to assess the effect of RFA vs PFA on CTI ablation time and total procedure time while controlling for each covariate. Procedure times were



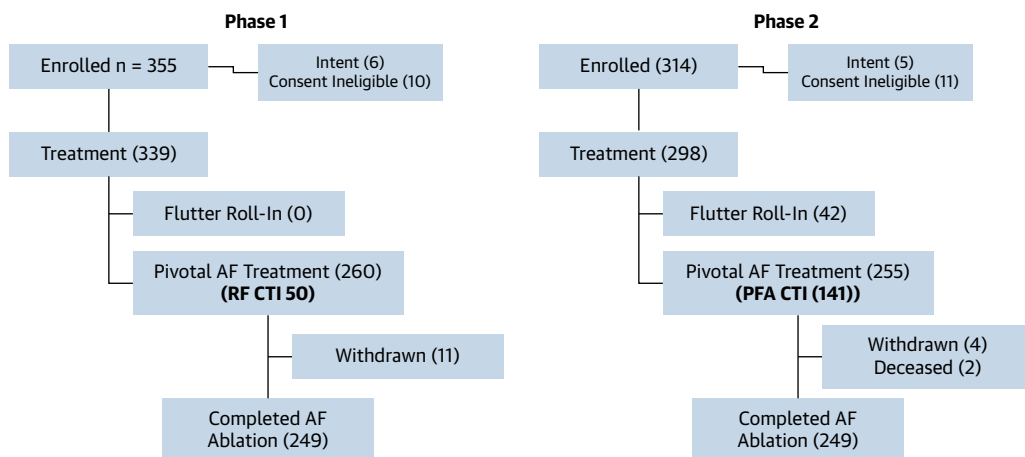
normalized by means of a natural log-transformation before regression. Analyses were conducted with SAS version 9.4 (SAS Institute Software Co).

RESULTS

PRECLINICAL STUDY. Eighteen swine were studied in the 6 randomized groups: 3 field strengths (1.4 kV, 1.7 kV and 2.0 kV) and at each voltage 1, 2, or 3 applications at each site. Thus, 2 animals were studied at each voltage and application number, with a median of 6 lesions delivered per animal for a total of 112 lesions analyzed. Histopathologic and gross measurements corresponded well, with both measurement methods finding similar trends in depth and surface width (Supplemental Figure 1). When considering potential tissue deformation during dehydration, microtoming, and staining, the differences between methods were small.

Lesion depth and surface width at different energy levels and application numbers are shown in Figure 4. Combining data for the number of applications delivered, the depth increased with higher field strength (1.4 kV vs 1.7 kV vs 2.0 kV, respectively: 3.3 ± 1.1 , 3.6 ± 0.7 , and 4.2 ± 0.8 mm; ANOVA $P < 0.0001$). Examination of the number of applications per site showed that depth started to plateau with multiple applications: At 1.4 kV there was a progressive increase in lesion depth with more applications, whereas at 1.7 and 2.0 kV, lesion depth plateaued at ~ 3.8 and 4.5 mm, respectively, after 2 pulsed field applications. Lesion width progressively increased with increasing voltage and number of applications. Lesions modeled to the best estimates for dimensions for each treatment are presented in Supplemental Figure 2. A previous detailed study of CTI morphology with the use of computed tomography found that the mean CTI

FIGURE 5 Consort Diagrams for Phase 1 and Phase 2 Advantage AF Studies



Abbreviations as in [Figure 1](#).

tissue thickness was 4.3 ± 1.5 mm in mid-systole, and a thick eustachian ridge >4 mm was seen in 24% of patients.¹⁷ Balancing procedure efficiency with lesion depth, and recognizing that lesion width >8 mm was less important to achieving CTI block, we determined the recommended applications of 2 applications/site at 2.0 kV in the clinical study. In the lesion volume modeled in [Supplemental Figure 2](#), adding beyond 2 pulsed field lesions per site at 2 kV added minimal lesion depth, whereas the added lesion width was considered to be unnecessary for achieving or maintaining CTI block.

CLINICAL STUDY. ADVANTAGE AF enrolled a total of 637 patients from February 2023 to March 2024. In phase 1, 50 patients received standard RF CTI ablation, and in phase 2, 141 patients received CTI ablation with the linear PFA catheter ([Figure 5](#)). Specific AFL ablation times were recorded in 48 of 50 phase I RF patients (2 patients omitted owing to redo of previous CTI ablation) and in all 141 phase 2 PFA patients. For the 42 roll-in PFA CTI ablation patients in phase 2, no safety events occurred, including no clinical evidence of CA spasm or hemolysis.

Demographics. Although patients in the 2 cohorts were generally similar, phase 2 PFA-CTI patients had significantly higher rates of heart failure and hypertension, and significantly lower rates of sleep apnea compared with phase 1 RFA-CTI patients ([Table 1](#)). More phase 2 PFA-CTI patients underwent CTI ablation compared with phase 1 RFA-CTI patients (55% [141 of 255] vs 19% [50 of 260]; $P < 0.005$).

Procedural details. Phase 2 PFA-CTI cases used EAM more frequently than phase 1 RFA-CTI cases (88.7% vs 68.0%; $P < 0.0005$). Patients in phase 2 undergoing CTI ablation with PFA had significantly shorter total procedure time (PVI + PWA + CTI) than those in phase I undergoing CTI ablation with RFA (median 97 min [Q1-Q3: 79-122 min] vs 118 min [Q1-Q3: 96-144 min]; $P = 0.005$) ([Figure 6](#)) and shorter total fluoroscopy time (16 min [Q1-Q3: 8-23 min] vs 22 min [Q1-Q3: 17-31 min]; $P < 0.001$). Elapsed CTI times showed that PFA-CTI patients had significantly shorter CTI ablation times than RFA-CTI patients (5 min [Q1-Q3: 4-8 min] vs 14 min [Q1-Q3: 9-19 min]; $P < 0.001$) ([Figure 6](#)). The number of PFA applications were similar for PVI and PWA across both cohorts. The total number of applications for CTI was similar between PFA and RFA ([Table 2](#), [Figure 6](#)), though it should be noted that each PFA lesion is delivered in 2.5 seconds compared with ~ 30 seconds for RFA. Acute bidirectional CTI conduction block was similar with PFA and RFA (139 of 141 [98.6%] vs 50 of 50 [100%]; $P = 1.00$). Examination of the variance of number of applications ($F = 8.0$; $P < 0.001$) showed that patients receiving PFA-CTI had significantly lower patient-to-patient variability ([Figure 6](#)).

Most patients in the PFA-CTI arm (140, 99.3%) received the 3 mg NTG pretreatment, and 65 (46.1%) patients received an additional 2 mg bolus during the CTI ablation. There was no relationship between duration of CTI ablation and total NTG dose ([Figure 7](#)). In addition, most (138, 97.9%) received a vasopressor before the CTI ablation, usually

TABLE 1 Baseline Characteristics of the RFA and PFA Groups

	RFA-CTI (n = 50)	PFA-CTI (n = 141)	P Value
Age, y	65.0 ± 10.0	66.3 ± 8.9	0.39
Female	10 (20.0)	44 (31.2)	0.13
Body mass index, kg/m ²	30.2 ± 5.1	30.8 ± 5.1	0.44
Comorbidities			
CAD	12 (24.0)	30 (21.3)	0.69
Congestive HF (NYHA functional class I or II)	13 (26.0)	73 (51.8)	0.002
Diabetes	8 (16.0)	30 (21.3)	0.42
Dyslipidemia	20 (40.0)	59 (41.8)	0.82
Hypertension	23 (46.0)	101 (71.6)	0.001
Sleep apnea	18 (36.0)	23 (16.3)	0.004
Stroke	3 (6.0)	3 (2.1)	0.19
Transient ischemic attack	2 (4.0)	2 (1.4)	0.28
Acute myocardial infarction	2 (4.0)	4 (2.8)	0.65
CHA ₂ DS ₂ -VAsC score	2.0 ± 1.5	2.4 ± 1.4	0.078
LA diameter, cm	4.3 ± 0.5	4.3 ± 0.7	0.81
LA volume, mL	60.0 ± 24.6	67.2 ± 18.1	0.51
LVEF, %	57.4 ± 6.5	56.2 ± 6.9	0.28
Years since AF diagnosis	3 ± 4	3 ± 4	0.99
Antiarrhythmic drugs at baseline			
Amiodarone	12 (24.0)	42 (29.8)	0.43
Dofetilide	4 (8.0)	12 (8.5)	1.00
Dronedarone	5 (10.0)	6 (4.3)	0.16
Sotalol	3 (6.0)	17 (12.1)	0.29
Flecainide	6 (12.0)	10 (7.1)	0.28
Propafenone	2 (4.0)	3 (2.1)	0.61
Anticoagulants at baseline			
DOAC	50 (100)	141 (100)	1.00
VKA	0 (0.0)	0 (0.0)	1.00
CTI ablation reason			<0.001
History of CTI-dependent flutter	31 (62.0)	80 (56.7)	
CTI ablation indicated for patient welfare	3 (6.0)	44 (31.2)	
CTI-dependent flutter manifest during the procedure	16 (32.0)	17 (12.1)	

Values are mean ± SD or n (%).

CAD = coronary artery disease; CTI = cavotricuspid isthmus ablation; DOAC = direct oral anticoagulant; HF = heart failure; LA = left atrial; LVEF = left ventricular ejection fraction; PFA = pulsed field ablation; RFA = radiofrequency ablation; VKA = vitamin K antagonist.

phenylephrine (134, 97.1%) at a median dose of 400 µg, administered either before ablation (68, 49.3%) or both before and during ablation (63, 45.7%).

It is interesting that phase 2 of ADVANTAGE AF had a higher percentage of CTI ablations performed than phase 1. Both patient cohorts had similar presenting rhythms at presentation, but there were differences in the reason for CTI ablation (Table 1). The PFA-CTI group in phase 2 appears to have had a higher rate of CTI attributed to patient welfare^{44/141} compared to RF-CTI,^{3/50} whereas RF-CTI patients were more likely to have received CTI because flutter manifested during the index procedure (16/50), compared with PFA-CTI (17/141). For full

presenting rhythm and CTI ablation reasoning, see Supplemental Table 3.

Efficacy and safety. At 90 days, the per-protocol CTI effectiveness rate for PFA was 98.6%. There were 2 acute procedural failures with PFA: 1) 1 patient required completion of the CTI line with RFA after a technical issue, namely, an accidental cable disconnection led to an unrecognized console error; and 2) bidirectional conduction block was not confirmed in a patient owing to ongoing AF. The 90-day effectiveness rate for RFA was 100% ($P = 1.00$). No clinical manifestations of CA spasm occurred during or after ablation in either group.

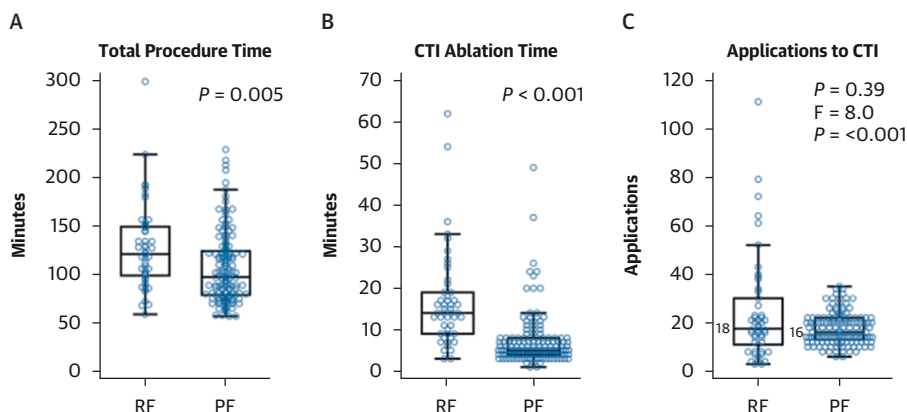
The 360-day freedom from recurrent typical AFL was not significantly different between PFA and RFA groups (97.9 vs 98.0%; $P = 0.95$, 8). Three patients who received PFA-CTI required re-ablation at days 131, 161, and 259 for recurrence of typical AFL. One patient who received RF CTI required re-ablation of the CTI at day 181 for recurrence of typical AFL.

For context, across all 513 pivotal treatment patients in phase 1 and phase 2 of ADVANTAGE AF, occurrence of any AFL during 1-year follow-up was 10.3% (10.0% in phase 1, 10.7% in phase 2). Of 53 total patients with AFL recurrence, 33 also had AF recurrence (73.1% in phase 1, 51.9% in phase 2), 6 had recurrence of both AFL and atrial tachycardia without AF (0.8% in phase 1, 1.6% in phase 2), 14 had only recurrence of AFL (1.9% in phase 1, 3.6% in phase 2), and 10 had only recurrence of atrial tachycardia (2.7% in phase 1, 1.2% in phase 2).

At 90 days, the per-protocol safety event rate was not different between the PFA and RFA groups (2.1% [3 of 141] vs 2.0% [1 of 50]; $P = 0.96$). Adverse events included pulmonary edema ($n = 1$) in the RFA group, and stroke ($n = 1$), tamponade ($n = 1$), and death ($n = 1$) in the PFA group; no clinical sequelae of hemolysis or CA spasm were reported. As previously reported,¹⁵ the mortality event occurred 24 days after the index procedure and was adjudicated to have been secondary to underlying cardiovascular disease. No additional safety events occurred in these patients over the remainder of the 360-day follow-up of the trial (Figure 8). Safety data have been previously reported on the roll-in patients.¹⁵ Notably, there were no instances of CA spasm in roll-in patients and only a single acute kidney injury, which was adjudicated to be likely secondary to use of a proton pump inhibitor.

Sensitivity analysis. To account for the possibility of bias due to demographic differences between the 2 patient cohorts, we conducted a sensitivity analysis to examine the contributions of sex, congestive heart failure, hypertension, sleep apnea, CHA₂DS₂-VAsC

FIGURE 6 Patient Level Procedure and CTI Ablation Times



Total procedure time (A) and CTI ablation time (B) were both significantly shorter with PF compared to RF ablation. There was no difference in number of CTI lesions delivered between groups (C), although PF lesions are shorter (2.5 seconds). The variation in CTI ablation time and # of CTI lesions was also significantly less with PFA ($F = 8.0$, $p < 0.001$). Each dot represents an individual subject time point. One time point at 140 minutes with PFA in plot B not shown.

score, reason for CTI ablation (clinically indicated vs prophylactic), transseptal puncture type (single vs dual), and use of EAM for PWA. Separate linear regression models were used to assess the effect of RFA vs PFA on CTI ablation time and total procedure time while controlling for each covariate. Procedure times were normalized by means of a natural log-transformation before regression.

Ablation modality (RFA vs PFA) remained a significant predictor of both CTI ablation time and total procedure time after adjustment for each covariate, with P values remaining <0.0001 and <0.01 , respectively. Dual transseptal access and no use of mapping for PWA were also significant predictors of shorter total procedure time ($P = 0.02$ and $P = 0.01$, respectively).

Because more patients in phase 2 used EAM than in phase 1 (Table 2), it is possible that this contributed to the shorter procedure time. We therefore performed a sensitivity analysis comparing total procedure time only in patients where EAM was used (34 of 50 in phase 1 and 124 of 140 in phase 2). One phase 2 patient was excluded because of tamponade/perforation leading to an aborted procedure. In the EAM subgroup, the total procedure time remained significantly shorter in phase 2 using PFA for CTI compared with phase I using RFA for CTI (95.5 min [Q1-Q3: 77.5-125.5 min] vs 114.5 min [Q1-Q3: 91-149 min]; $P = 0.020$).

DISCUSSION

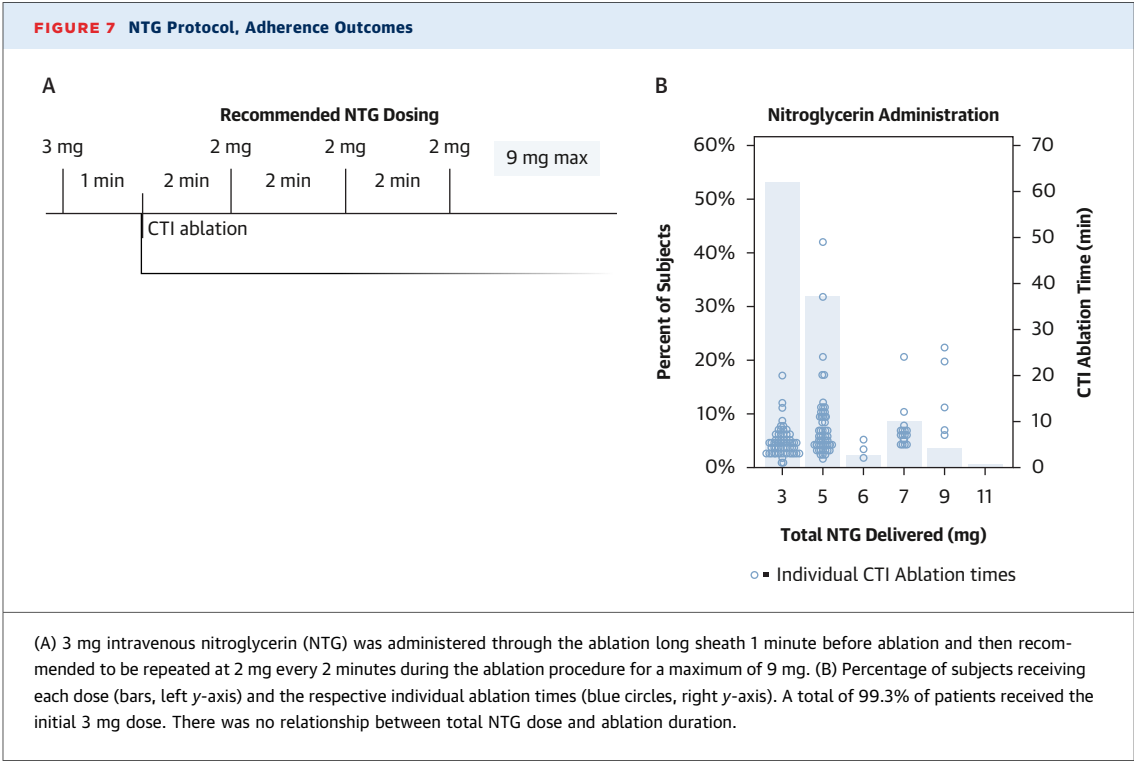
The major findings of this study are that: 1) CTI ablation with a novel linear bipolar PFA catheter was associated with similar effectiveness and shorter

TABLE 2 Procedural Details of the RFA and PFA Groups

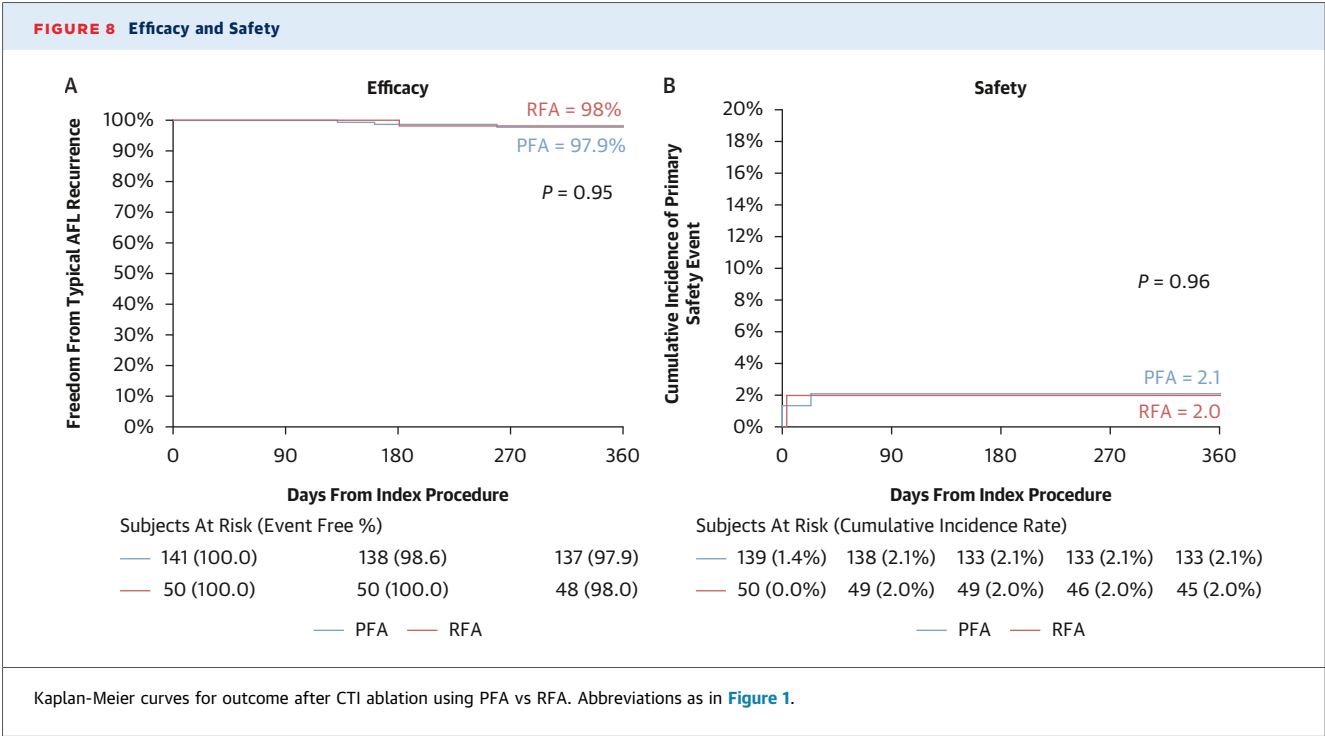
	RFA-CTI (n = 50)	PFA-CTI (n = 141)	P Value
Acute success	50 (100)	139 (98.6)	1.00
Procedure times, min			
Total procedure	121 (99-149)	98 (79-124) ^a	0.001
LA dwell	54 (44-77)	54 (42-73)	0.41
Fluoroscopy	24 (17-32)	16 (8-23)	<0.001
PV isolation	21 (16-32)	18 (15-26)	0.014
PW isolation	20 (8-27)	9 (7-19)	0.017
CTI ablation	14 (9-19)	5 (4- 8)	<0.001
No. of energy applications			
PVI	47 (38-53)	43 (38-50)	0.13
PWI	32 (26-40)	32 (24-40)	0.68
CTI	18 (11-30)	16 (13-22)	0.39
Electroanatomic mapping system use (for PW)	34 (68.0)	125 (88.7)	<0.001
CTI	-	-	
Total nitroglycerin, mg	-	4 ± 2	N/A

Values are n (%), median (Q1-Q3), or mean ± SD. ^aExcluding 1 subject where CTI and PVI+PWA were performed on separate dates.

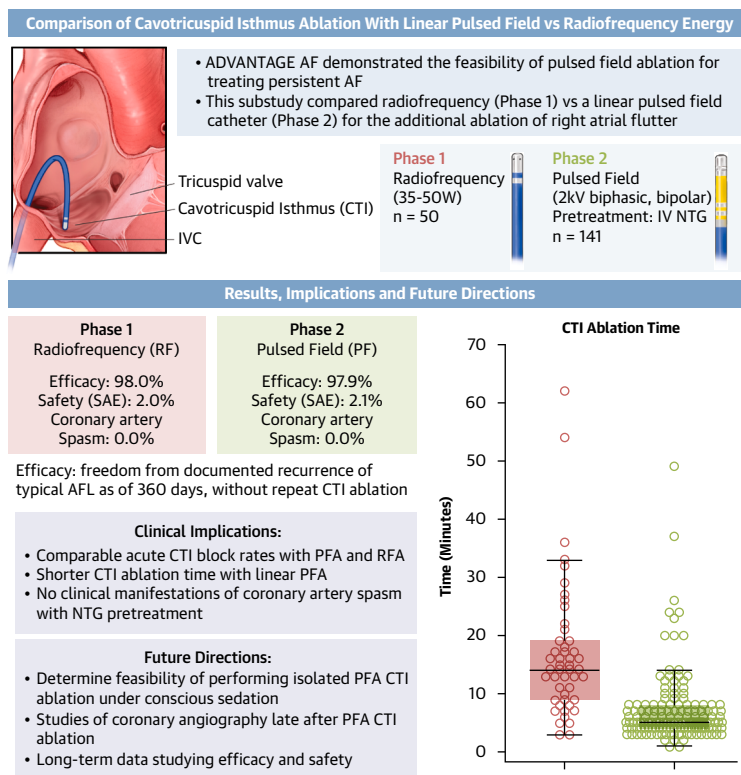
PV = pulmonary vein; PVI = pulmonary vein isolation; PW = posterior wall; PWI = posterior wall isolation; other abbreviations as in Table 1.



procedure duration compared with RFA ([Central Illustration](#)); and 2) there were no acute clinical manifestations suggestive of significant CA spasm during PFA ablation of the CTI after NTG pretreatment. CTI ablation has been performed safely and effectively with RFA for >30 years, but thick pectinate muscles and intra-isthmus pouches can add significant challenges, sometimes necessitating



CENTRAL ILLUSTRATION Cavotricuspid Isthmus Ablation With Pulsed Field Energy Is Effective, Safe, and More Efficient Than Radiofrequency Energy



Gerstenfeld EP, et al. JACC. 2026;87(14):1764-1778.

Examples of the pulsed field ablation (PFA) and typical irrigated RF ablation catheters are shown. Overall, PFA had similar efficacy and safety to radiofrequency (RF) ablation, with more efficient and less variable procedure times. There was no evidence of coronary artery spasm after the recommended nitroglycerin (NTG) pretreatment. CTI = cavotricuspid isthmus; IV = intravenous.

numerous RF lesions and often adding significant time to an AF ablation procedure. In addition, pericardial symptoms such as chest pain and dyspnea, which have been largely mitigated with PFA, may occur when RFA of the CTI is added to PFA. PFA has been shown to be safe and efficient for AF ablation and is gaining widespread acceptance; however, treatment for AFL with PFA has been limited to off-label adoption of large-footprint catheters for CTI ablation, with risk of excessive tissue destruction or CA injury. The present study demonstrates that PFA of the CTI with the use of a linear PFA catheter is associated with success at 1 year similar to RFA, with significantly shorter (median: 5 min) and more predictable [Q1-Q3: 4-8 min] total procedure duration when added to AF ablation. One might wonder why PFA was associated with a 21 minutes shorter total

procedure time, when the actual CTI ablation time was shortened by only a mean of 9 minutes. This is likely because switching from PFA to RFA during the procedure requires setting up another generator, infusion pump, and tubing, whereas the focal PFA catheter uses the same ablation generator and no saline infusion, dramatically shortening the overall procedure time.

Concerns about CA spasm from PFA were initially raised during ablation of the lateral mitral isthmus, where the proximity to the left circumflex CA is well recognized.¹² Preclinical studies then found that epicardial or endocardial PFA ablation within 6.5 mm of a CA can lead to spasm.¹⁸⁻²⁰ Although reports of CA injury have been rarely reported with RF ablation of the CTI,^{21,22} many operators may not have previously appreciated the close proximity of the right CA to the

CTI.²³ Intravenous (IV) NTG pretreatment has been shown to greatly reduce the risk of subclinical acute CA spasm from PFA.^{13,16}

Based on preclinical and clinical data, a protocol for NTG treatment before and during PFA ablation of the CTI was developed.¹⁶ This protocol, used in 141 patients in the present study with no clinical manifestations of CA spasm, provides the first reproducible coronary-safety strategy for PFA of the CTI. Although this dose of IV NTG may be perceived as being quite high, pretreatment with vasopressor bolus led to no adverse effect in this multicenter study. It should also be noted that although 99% of patients received the initial 3 mg IV NTG bolus, only 46% received the additional 2 mg bolus during the CTI ablation. This may be because either: 1) CTI ablation was completed before the next dose was indicated; or 2) operators felt that once the catheter was withdrawn proximal to the annular portion of the CTI where proximity to the right CA was of greatest concern, additional NTG was less useful. However, it should be recognized that subclinical right CA spasm may still occur, and caution should be used with PFA ablation of the CTI. In a recent study using monopolar PFA for CTI ablation, clinical vasospasm was noted in 5% of patients¹¹; however, patients were pretreated with only 100-400 μ g IV NTG, which is likely insufficient. Furthermore, Stojadinović et al, in a randomized trial examining flower vs basket pose of the pentaspline PFA catheter for CTI ablation, reported transient ST-segment elevations in only a single patient (1/178).²⁴ Notably, that patient had not received the 2 prophylactic 0.3 mg sublingual NTG sprays prescribed in the protocol, and the event did not require further treatment or prolong hospitalization.²⁴ Recently, Turagam et al, for MANIFEST US, reported very low rates of spasm (0.1%), but also observed that rates in a subpopulation of Pacific Islanders was much higher; suggesting that there is much yet to be learned about acute spasm during ablation procedures.²⁵ Finally, it should be emphasized that there are little data on the utility of the NTG protocol in patients with known multivessel CA disease.

Beyond the acute effect of PFA on CAs, and its mitigation with NTG pretreatment, there are still open questions as to the chronic effect of PFA vs RFA on CAs. In one study in which invasive coronary angiography was performed a median 11 months after CTI ablation with the pentaspline PFA catheter, acute CA spasm did not translate to long-term coronary lesions.¹⁸ But in another study, using high-resolution optical coherence tomography, there was evidence of

mild CA injury 3 months after ablation of the CTI using the pentaspline PFA catheter.²⁶ However, the latter study did not have an RFA comparator. Direct comparisons of PFA vs RFA have been performed in 2 preclinical studies, with both finding that chronic CA lesions—focal medial fibrosis and intimal hyperplasia—were more common with RFA than PFA.²⁷⁻²⁹ Of note, frank CA stenosis was not observed with either energy modality. These data, together with the present study, argue that PFA of the CTI with NTG pretreatment is safe and effective over the medium term. However, we recognize that longer-term follow-up studies are needed to confirm safety.

The PFA-CTI protocol in this study included 2 applications of 2 kV at each site as the catheter was withdrawn from the tricuspid annulus to the inferior vena cava. One might wonder if one application per site could result in an even shorter procedure, or whether more applications per site could be beneficial. A detailed study examining CTI morphology by means of computed tomography found the mean CTI tissue thickness to be 4.3 ± 1.5 mm in mid-systole.¹⁷ The two 2 kV PFA applications per site used in the present study were based on our preclinical testing in the swine ventricle with the use of the linear PFA catheter (Figure 4). The design of this catheter should also be considered, because the linear design facilitates some overlap of applications when the catheter is parallel to tissue, which may facilitate achieving complete CTI block. Notably, the present catheter used a bipolar PFA waveform. While unipolar waveforms may be expected to have greater lesion depth, the risks of additional skeletal muscle stimulation and CA spasm or injury may be drawbacks of a unipolar waveform.^{30,31} The complexity of varying CTI anatomies would make a rigorous clinical study of PFA dosing for CTI challenging, so preclinical lesion characterization was used to inform the present protocol.

Efficacy (98%) and safety (98%) for PFA of the CTI was quite high in this study. A recent registry of >300 patients with AFL reported that on the use of the pentaspline PFA catheter to treat right- and left-side AFL.³² They reported high acute efficacy (99.5%), safety (99%), and short-term success rates (98.5%) when PFA was used for CTI ablation.³² While showing that a large-footprint PFA catheter can be used for adjunctive ablations, the present study also calls attention to the need for smaller linear PFA catheters in patients with small atria, as well as to mitigate the possible risk of CA spasm. It should therefore be emphasized that the excellent safety outcome reported in this study should apply only to linear bipolar PFA catheters; it is not clear that other

PFA waveforms and catheter configurations would have similar results.

STUDY LIMITATIONS. This was a post hoc analysis comparing data from 2 separate phases of a single trial. Although this is the largest series comparing PFA and RFA of the CTI, less common adverse events cannot be excluded. Notably, a recent registry on a larger series of consecutive patients that received adjunctive PFA ablations with the pentaspline catheter reported similarly low adverse event rates.³² The fact that the present analysis is a post hoc comparison rather than a randomized controlled trial leaves the possibility that recruitment or treatment bias could affect our results. The dose of IV NTG was large and typically required pretreatment with vasopressor agents. Whether lower doses can be used requires further study. Detailed CA evaluation before enrollment was not performed, which is typical of patients referred for AF ablation. However, it is possible that clinical manifestations of CA spasm, should they occur, may be more severe in patients with multi-vessel CA disease. Because patients in the study were typically under general anesthesia for concomitant AF ablation, results cannot be extrapolated to patients undergoing isolated CTI ablation under conscious sedation, particularly as significant diaphragmatic or skeletal muscle stimulation may occur with the use of PFA.

CONCLUSIONS

Although it was not a randomized comparison, the use of bipolar linear PFA was associated with similar efficacy and safety and superior efficiency compared with RFA. This may extend electroporation from the pulmonary veins to linear substrates, and allow a single nonthermal ablation platform for treatment of AF and AFL.

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KEY WORDS ablation, cavotricuspid, coronary artery, flutter, pulsed field

APPENDIX For supplemental tables and figures, as well as statistical analysis plans and a clinical investigation plan, please see the online version of this paper.